ACME DC Charger Controller

Reference Manual

Document ACME-RM-042, Rev 1.2

Revision history

Rev 1.2 adds the thermal derating rule and corrects the contactor ratings.

Rev 1.1 adds Appendix A.

1 Introduction

The controller supervises the power stage and the vehicle interface. It samples the bus voltage, the output current and three temperature sensors every 10 ms and reports the values over the internal CAN bus. All limits in this chapter apply over the full operating temperature range unless a different condition is stated in the table.

1.1 Scope

This manual describes the 60 kW DC charger controller board, hardware revision C. It covers electrical limits, the charging sequence and the register map.

1.2 Terms

- EVSE: electric vehicle supply equipment.
- CCS: combined charging system.

The term inter-
rupt latency means the time from the fault to the opening command.

2 Electrical characteristics

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2.1 Supply

The logic supply is 3.3 V. The contactor coil supply is 12 V.

2.1.1 Limits

Parameter	Min	Typ	Max	Unit
Logic supply voltage	3.0	3.3	3.6	V
Coil supply voltage	10.8	12.0	13.2	V
Bus voltage	200	400	480	V
Output current	0		200	A

2.2 Contactor

The output contactor is rated for 200 A. It is controlled by the control board, which receives its commands from the MCU.

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3 Charging sequence

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3.1 Thermal derating

If the temperature exceeds 85 °C, the controller shall reduce the charging current unless emergency mode is enabled.

If the output current exceeds 180 A for more than 2 s, the controller shall open the contactor.

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3.2 Shutdown

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Appendix A Register map

A.1 Status register

Bit	Name	Access	Reset	Description
0	READY	R	0	Controller is ready to charge
1	FAULT	R	0	A fault is latched
2	DERATE	R	0	Thermal derating is active
7:3	Reserved	R	0	Read as zero

Writing to a reserved bit has no effect.